Anchor Point Risk

SPECIALIST RATING SERVICES

The LGD tail extension model: what it does and how it works

Methodology, design rationale and evidence, for model validation | 20 October 2026

The risk suite measures loss given default (LGD) from recovery triangles. For each month since default, a triangle records the share of the defaulted balance that came back as cash. Those triangles are truncated. The longest-running cohorts are thin, and the last months of each curve rest on a handful of accounts, so the observed tail is either missing or too noisy to use as it stands.

The tail extension model keeps the part of each recovery curve that is backed by enough exposure. It replaces the rest with a fitted decay curve, runs that curve out to a fixed horizon, and recomputes LGD. The model is implemented twice. The **risk suite app** (the `hazard_ext` Python engine behind the web application) runs it at scale across projects, debug files and scenarios. The **validation workbook** (`LGD_Validation_vb11_ReportDate_Optimised1.xlsb`) rebuilds the same calculation in Excel formulas so that every number can be traced by hand.

This document is written for model validation. It first sets out the model's purpose and scope. It then describes the calculation once, in the order it runs, with a note at each stage on where it sits in the app and in the workbook. Worked examples use the category 11 debug file (Lifetime event, report date July 2026). The later sections give the reasoning behind each design choice, the evidence for the choice of tail shape, sensitivity and materiality on two contrasting books, the reconciliation of the two implementations, and the model's known weaknesses, controls and monitoring.

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1 The model at a glance

A loan that defaults at month t_s of its cohort's life is said to default at *TermStep* t_s. Its recoveries arrive in later *buckets* $b \geq t_s$, where bucket b is month b of the cohort's life. The RecoveryPct $R(t_s, b)$ is the cash received in bucket b as a share of the balance at default. LGD is one minus the present value of those recoveries, discounted to the default date:

$$\text{LGD}(t_s) \;=\; 1 - \frac{\sum_{b \geq t_s} R(t_s, b)\, v^{\,b}}{v^{\,t_s - 1}}, \qquad v = (1+r)^{-1/12}.$$

The risk suite's own LGD stops the sum where the data stop. The tail extension changes only what happens after the last *credible* bucket. Up to that point the ob-

served recoveries are kept exactly; after it they are replaced by a fitted curve out to MaxBucket (420 months by default). The model runs in seven stages, and both implementations follow the same order.

Stage	What it does	App (`engine/core.py`)	Workbook
1 Data selection	Report date; which cohorts are used	`parse.resolve_vintages`	Config, Runoff_Input F–G
2 Observed triangles	Exposure and RecoveryPct by TermStep and bucket	`parse.vintage_triangles`	Obs_E, Obs_R
3 Credibility	Last credible bucket and anchor window	`_credible_window`	Tail_Fit B–F
4 Decay fit	$\lambda, \gamma, \mu, \sigma$ on the reference row	`_fit_decay`, `fit.py`	Fit_OLS, Fit_LM, RunLMFit
5 Extension	Level per TermStep; fitted tail beyond the credible part	`_scales`, `_extend`	Tail_Fit H–J, Shapes
6 LGD per TermStep	PV of observed plus fitted recoveries	`_row_lgds`, `_results`	Tail_Fit N–S, Results
7 LGD to Target	LGD for every TermStep to the target	`_lgd_to_target`	Base_Row, LGD_Target

Example 1 — The headline result, category 11 at July 2026

With the default settings (exponential shape, Levenberg–Marquardt fit, MinExposure R100,000,000, window 12, FitStart 24, MaxBucket 420), the exposure-weighted LGD is **0.9816185** in both the app and the workbook. The risk suite's original LGD is **0.9816104**.
The change in PV recoveries is -8.29×10^{-6} of exposure. At TermStep 1 (R27.45bn of defaulted exposure) the extended LGD is 0.9568860, against the original 0.9568846.

2 Purpose, scope and intended use

2.1 Purpose

The risk suite's LGD for a defaulted exposure is the present value of the recoveries observed on its recovery curve. Where the curve is long and well populated, that is a sound estimate. Where it is truncated, or where its last months rest on a few accounts, the estimate either omits recoveries that are still to come or gives weight to a handful of idiosyncratic repayments. The tail extension model exists to correct that one weakness. It leaves the credible part of the observed curve untouched and replaces only the part that is not credible.

2.2 What the model produces

- **LGD by TermStep**, i.e. by months in default at the point of measurement, for each of the three tail shapes and for the selected one.
- **The exposure-weighted LGD** across TermSteps, with the risk suite's original LGD beside it and the difference in PV recoveries.
- **LGD within a horizon**: the same measure counting only recoveries in the next 12 months (short horizon) or 120 months (valuation horizon). These show how much of the recovery is near-term and how much depends on the tail.
- **LGD to the target TermStep**, extending the LGD table past the last observed TermStep by rolling a base curve forward.

The outputs are designed to stand in for the risk suite's LGD where its tail is not credible. How they enter the expected credit loss calculation (for example, which TermStep an account is mapped to, and whether the horizon measures are used) is set by the ECL methodology and is outside this document.

2.3 Scope and boundaries

The model works on one event block of one debug file at a time. It uses the run-off balances and the file's discount factor, and nothing else. In particular:

- **A recovery is any fall in the outstanding balance** from one month to the next. The model therefore relies on the run-off file's balances being reduced by cash received. Any write-off, interest or fee posting that changes the balance flows straight into the recovery rate.
- **It is historical.** It measures past recovery behaviour of the selected cohorts. It contains no forward-looking macroeconomic adjustment; any such adjustment sits outside it.
- **It does not model** collateral values, recovery costs, cures or re-defaults separately. Whatever of these is in the balance history is in the recovery curve; whatever is not, is not.
- **It does not change the observed credible data.** Every observed recovery up to the last credible bucket enters the LGD exactly as recorded.

3 Inputs

3.1 The debug zip

Every risk suite run writes a debug zip, and the model needs three files from it.

- `runoff_triangle.csv` holds the raw balances: one row per (EventType, CohortDate, Bucket) with ExposureAmount and AccountCount. This is what the model calculates from. For category 11 it has 6,695 rows over 59 cohorts, June 2001 to July 2026.
- `lgd_recovery.csv` holds the risk suite's own triangle and LGD, one row per (EventType, TermStep, bucket). The model uses it as the reference that the rebuilt triangle must tie to, and as the source of the discount factor. For category 11 it has 45,753 rows over 302 TermSteps.
- `debug.json` holds the run parameters. `LgdMaxDate` is the report date (31 July 2026) and `InterestRate` the discount rate (0.258). Both can also be derived from the two CSVs, which is how the model gets them; the JSON is the check.

The debug files carry three event blocks (EventType): Lifetime, LifetimeSingle and TwelveMonthSingle. In both files examined here the three blocks are identical row for row, in the run-off data and in the recovery triangle, so they give the same LGD. The app detects this, and the workbook holds the Lifetime block only. Where the blocks differ, each is modelled separately. In the app, a debug zip is uploaded to a project as a dataset, and one project can hold many zips (one per category or portfolio). In the workbook, the two CSVs are pasted onto **Runoff_Input** and **Input** unchanged and in their original sort order, and the event type is chosen on Control.

3.2 Parameters

The same parameters drive both implementations. In the app they form a *scenario*, which can carry per-zip overrides; in the workbook they are the yellow cells on Control.

App parameter	Control	Meaning	Default
`event_type`	C11	event block of the debug file	Lifetime
`vintage_years`	C12–C13	keep cohorts from the last N years to the report date	all
`report_date`	C14	month that "last N years" counts back from	from data
`vintage_start`	—	keep cohorts from a given month (app only)	all

App parameter	Control	Meaning	Default
`min_exposure`, `_mode`	C16–C17	credibility cut, rand or % of opening exposure	R100m
`window`	C18	anchor window W, in buckets	12
`fit_start`	C19	first bucket of the fit window	24
`ref_ts`	C20	TermStep the shapes are fitted on	1
`method`	C22	1 exponential, 2 power law, 3 log-normal	1
`fit_solver`	C23	LM (weighted) or OLS (log-linear)	LM
`floor`	C24	minimum RecoveryPct on the fitted tail	0
`lambda_override` etc.	C25–C28	fix $\lambda, \gamma, \mu, \sigma$ instead of fitting	fitted
`rate`	C30	discount rate p.a.	implied by file
`target_ts`	C31	last TermStep in the LGD-to-Target table	300
`max_bucket`	C32	horizon of the extension	420
`horizon`, `horizon2`	C33–C34	short and valuation horizons (months)	12, 120
`base_ts`	C35	row rolled forward for derived LGD	1
`last_ts`	C36	last TermStep used as observed	from data

4 Stage 1 — Report date and data selection

The report date is the last month the data can see. A cohort that started in month m and has been followed to bucket K was last observed in month $m + K$. The report month is therefore the largest $m + K$ across the cohorts of the event. A complete run-off file gives the same answer from every cohort.

> **Example 2 — Deriving the report date for category 11**
>
> The oldest cohort, June 2001, runs to bucket 301. June 2001 plus 301 months is July 2026. The newest cohort, July 2026, has only bucket 0, which gives July 2026 again. That agrees with `LgdMaxDate` of 31 July 2026 in `debug.json`.

The report date can be overridden, which is useful when the run-off file reaches past the reporting date being prepared.

4.1 Last N years counts back from the report date

"Last N years" keeps the cohorts that started within the N$\times$12 months ending at the report month. The first month kept is the report month less N years plus one month. The cohorts outside the window are dropped before any triangle is built, so every later stage sees only the selected book.

Example 3 — Five years on the VB22 book

The VB22 run-off covers 71 cohorts, December 2013 to September 2025, observed to July 2026. The report month is 202607, even though the newest cohort started in September 2025.
Five years back from July 2026 starts at **August 2021**, which keeps 34 of the 71 cohorts. With a report date of June 2026 the start moves to July 2021, which keeps 35 cohorts.

Counting back from the newest cohort

An earlier version counted the five years back from the newest *cohort* rather than the report date. On VB22 that put the start at October 2020, ten months too early.
The error grows with the gap between the newest cohort and the report date, so it bites hardest where origination has slowed. Both implementations now count back from the report date and report the report month with the result.

IFRS 9 — reasonable and supportable information (paraphrased)

IFRS 9.5.5.17 requires expected credit losses to reflect reasonable and supportable information about past events, current conditions and forecasts that is available at the reporting date. IFRS 9.B5.5.52 adds that historical loss information is adjusted for current observable data so it reflects present conditions.
A vintage window anchored to the report date is the mechanical expression of both: the last five years means the five years before the reporting date.

Implementation. In the app, `resolve_vintages` turns `report_date`, `vintage_years` or `vintage_start` into the cohorts kept. A filtered result carries a warning naming the window, the cohorts included and their share of exposure. In the workbook, `Config!ReportData` derives the report month, `StartMonth` applies the window, and the Keep flag in **Runoff_Input** column G marks each row in or out. The workbook offers "last N years" only; the app also accepts a start month.

5 Stage 2 — Observed triangles

For TermStep t_s and bucket $b \geq t_s$, the model takes the cohorts that still had a balance at bucket $t_s - 1$ and are observed at b. It divides their total cash in b by their total balance at default:

$$E(t_s, b) = \sum_c X_c(t_s - 1), \qquad R(t_s, b) = \frac{\sum_c \big(X_c(b-1) - X_c(b)\big)}{E(t_s, b)},$$

where $X_c(b)$ is cohort c's balance at bucket b. The sums run over kept cohorts with $X_c(t_s - 1) > 0$ whose last observed bucket is at least b. This is the risk suite's own definition, so on all vintages the rebuilt triangle equals the one in `lgd_recovery.csv`.

> **Example 4 — The first cells of TermStep 1 and 2**
>
> Every category 11 cohort observed to bucket 1 had R27,454,596,207.02 at default, so $E(1,1)$ = R27,454,596,207.02. The cash in bucket 1 is R23,714,303.30, so $R(1,1) = 0.00086376$.
> TermStep 2 starts from bucket 1 balances, so $E(2,2)$ = R27,430,881,903.72. That is $E(1,1)$ less the bucket 1 cash, as it must be when no cohort drops out between the two buckets.

The discount rate comes from the same file. The DiscountFactor at bucket 1 is 0.98105483, so the implied annual rate is $(1/0.98105483)^{12} - 1 = 25.8\%$, the `InterestRate` in `debug.json`. A typed rate overrides it.

5.1 Tie-out to the risk suite

Before anything is extended, the rebuilt triangle is tied to the file. Rebuilt exposure and cash match the file to within 10^{-11} (relative) for all 302 TermSteps. The *replica* LGD, the formula above on observed data only, matches the file's $1-$CumulativeSumPV to within 10^{-15}.

The risk suite also floors each TermStep's LGD at the largest final LGD of any earlier TermStep, so its LGD never falls as TermStep rises. The model reports the floor separately and does not apply it to the extended LGD. On category 11, 12 TermSteps carry a floor. At TermStep 2 the file's LGD of 0.9568846 is lifted from 0.9568775, a floor of 7.09×10^{-6}.

Implementation. The app sums cohort by cohort (`vintage_triangles`), which handles any run-off layout. The workbook gets the same sums with two SUMIFS per cell, using helper columns for each row's cohort month, last bucket and cash. That shortcut is exact only when each cohort's buckets run 0, 1, 2, ... without gaps and no balance returns from zero. The StructOK check tests both conditions on every row.

6 Stage 3 — Credibility and the anchor window

A recovery rate is only as reliable as the balance behind it. For each TermStep the *last credible bucket* is the last bucket where $E(t_s, b)$ is still at least MinExposure. MinExposure is set in rand or as a percentage of TermStep 1's opening exposure. If no bucket

clears the cut, the last observed bucket is used instead.

Observed recoveries up to and including the last credible bucket are kept. Everything after it is replaced. The *anchor window* is the last W credible buckets, and it sets the level of the replacement curve.

Example 5 — Why TermStep 1 stops at bucket 262

Under TermStep 1 the exposure behind the curve is R105,475,749 at bucket 262 and R77,921,457 at bucket 263. The first is above the R100,000,000 cut and the second below it, so 262 is the last credible bucket and the anchor window is buckets 251 to 262.
From bucket 263 to 301 the only cohort left is June 2001, which holds **one account**.

A long observed tail can rest on one account

Without the cut, a single account's repayments would set 39 months of the portfolio curve at TermStep 1. The cut is a judgement. On category 11, R100,000,000 is about 0.36% of the R27.45bn opening exposure.
A rand cut is also not scale-free. With a vintage filter the book is smaller, and the same rand cut can leave fewer credible buckets than the window needs; the app warns when this happens. A percentage cut scales with the book.

On category 11, from TermStep 253 onwards no bucket clears R100,000,000 at all, so those TermSteps keep all their observed data (to bucket 301) and are extended only beyond it.

7 Stage 4 — Fitting the decay shapes

The model offers three shapes for the tail:

$$s_{\exp}(b) = e^{-\lambda b}, \qquad s_{\text{pow}}(b) = b^{-\gamma}, \qquad s_{\ln}(b) = \frac{1}{b}\exp\left(-\frac{(\ln b - \mu)^2}{2\sigma^2}\right).$$

The exponential has a constant decay rate and a half-life of $\ln 2/\lambda$ months. The power law decays more slowly and has a heavier tail. The log-normal rises to a peak at $e^{\mu-\sigma^2}$ months and then falls, which suits books where recoveries build through legal collection before tailing off.

The parameters are fitted once, on the *reference TermStep* (default 1), and then shared by every TermStep. The fit window runs from FitStart (default 24) to the reference row's last credible bucket, using the buckets with $R > 0$. For category 11 that is buckets 24 to 262, 239 points.

7.1 The LM solver: weighted least squares on the recovery scale

The default solver minimises the weighted squared error on the recovery scale,

$$\min_{\theta} \sum_{b} w_b \left(c\, s(b;\theta) - R_b\right)^2, \qquad w_b \propto b.$$

The weights rise with the bucket, so the far tail, the part being extrapolated, counts most. For any trial θ the best level c has a closed form, so only the shape parameters are searched, by Levenberg–Marquardt from several starting points. The parameters are bounded: $\lambda \in [-10, 10]$, $\gamma \in [-50, 50]$, $\mu \in [-20, 30]$, $\sigma \in [0.02, 10]$. The level c is discarded; each TermStep gets its own level in Stage 5.

> **Example 6 — Fitted parameters, category 11**
>
> LM on buckets 24 to 262 gives $\lambda = 0.0184849$, a half-life of 37.5 months. It gives $\gamma = 1.3730$, $\mu = 3.9200$ and $\sigma = 0.8688$, so the log-normal peaks at 23.7 months with a median of 50.4 months.
> The app's "Forecast curve parameters" panel shows 0.0185, 1.373, 3.920 and 0.869.

7.2 The OLS solver: log-linear least squares

The alternative takes logs, which turns each shape into a straight line:

$$\lambda = -\text{slope}(\ln R \text{ on } b), \qquad \gamma = -\text{slope}(\ln R \text{ on } \ln b).$$

For the log-normal, $\ln R + \ln b$ is a quadratic in $\ln b$ with coefficients a_0, a_1, a_2, and then $\sigma^2 = -1/(2a_2)$ and $\mu = a_1\sigma^2$. OLS weights the early, large recoveries more than LM does, so its parameters differ. Only LM reconciles to the risk suite's curve fit.

7.3 When a shape is undefined

A fitted λ or $\gamma \le 0$ describes recoveries that grow forever, so the model treats that shape as undefined rather than extrapolating it. Under OLS, a log-normal with $a_2 \ge 0$ has no peak and is undefined in the same way; under LM, σ is bounded, so the log-normal can fail only if the fit does not converge. An undefined shape that is not the selected method gives a blank LGD and a warning. If the undefined shape is the selected method, the run stops with an error asking for a lower FitStart, an override or another method. Overrides fix a parameter at the user's value; under LM the other parameter of the log-normal is refitted around it.

Implementation. The app fits in `fit.py`. It starts from the OLS estimate and then from a fixed grid of starts, uses up to 500 iterations, and raises a warning when a fit fails or

a decay rises. The workbook does OLS in formulas on **Fit_OLS**, using SLOPE and a two-stage form of the quadratic fit. The LM fit is the `RunLMFit` macro (module `LM_Solver` v7), which writes $\lambda, \gamma, \mu, \sigma$ to **Fit_LM**. The macro uses its own grid of starts, but within the same bounds both solvers reach the same minimum. On category 11 the two agree to 3×10^{-11} on λ and 2×10^{-8} on γ.

The workbook's LM fit is a macro

The macro's results are values, not formulas, so they go stale when the inputs change. The workbook fingerprints the fit data. If the data no longer match the last run, Control shows "STALE — run RunLMFit" and the status turns to ATTENTION. A stale fit once showed $\mu = 1{,}559$ on the dashboard.
Version 7 of the macro also rejects any trial step outside the bounds, or one that would overflow. A step to $\lambda \approx -1.5$ had caused VBA runtime error 6.

8 Stage 5 — Level and extension

The fitted shape fixes how fast the tail decays. Its level is set separately for each TermStep, so that the curve matches that TermStep's own observed recoveries in total over its anchor window from D to E:

$$\text{scale}(t_s) = \frac{\sum_{b=D}^{E} R(t_s, b)}{\sum_{b=D}^{E} s(b)},$$

$$\tilde{R}(t_s, b) = \max\big(\text{floor},\ \text{scale}(t_s) \cdot s(b)\big) \quad \text{for } C < b \le \text{MaxBucket},$$

where C is the last credible bucket. Every TermStep's tail therefore decays at the same rate, but each joins smoothly onto its own recent recoveries.

Example 7 — The exponential level at TermStep 1

Over buckets 251 to 262 the observed recoveries sum to 0.00023443 and $e^{-0.0184849\,b}$ sums to 0.10493, so the scale is 0.00223407. The power and log-normal scales are 0.0396616 and 0.0289128; the app's panel shows the same three.
At bucket 263 the extended RecoveryPct is $0.00223407 \times e^{-0.0184849 \times 263} = 0.0000173$, in place of the single account's observed 0.0000263.

Implementation. The app builds the extended triangle for each shape as an array (`_extend`). The workbook does not build it: **Shapes** holds each shape by bucket and **Tail_Fit** holds the three scales per TermStep. The sums Stage 6 needs are then taken directly, row by row.

9 Stage 6 — LGD per TermStep

The extended LGD for each TermStep and shape is the observed PV to the last credible bucket plus the PV of the fitted tail:

$$\mathrm{LGD}_{\mathrm{shape}}(t_s) = 1 - \frac{\sum_{b=t_s}^{C} R(t_s,b)\, v^b + \sum_{b=C+1}^{\mathrm{MaxBucket}} \tilde{R}(t_s,b)\, v^b}{v^{\,t_s-1}}.$$

The selected method's LGD is reported alongside all three. Horizon LGDs apply the same formula with the sum cut off at $t_s + h - 1$, for the short horizon (12 months) and the valuation horizon (120 months).

Example 8 — LGD at TermStep 1, exponential

The observed PV to bucket 262 is 0.04311094. The fitted tail from bucket 263 to 420 adds 0.00000305, so

$$\mathrm{LGD} = 1 - (0.04311094 + 0.00000305) = 0.95688600.$$

The file's PV of recoveries is 0.04311542. The single account's buckets 263 to 301 contributed 4.47×10^{-6} of that; the fitted tail puts 3.05×10^{-6} in their place. PV recoveries fall by 1.42×10^{-6}, so LGD rises slightly.

The uplift can go either way, because the model *replaces* the non-credible observed tail rather than adding to it. At the early TermSteps of category 11 the surviving account repays faster than the portfolio curve implies, so the extension lowers recoveries. At the late TermSteps the curve is cut short by the end of the data, not by thin exposure, so the extension adds recoveries. At TermStep 295 PV recoveries rise from 0.00012 to 0.00052. Across the book the two net to -8.29×10^{-6} of exposure.

Portfolio figures weight each TermStep by its exposure at default, $E(t_s, t_s)$:

$$\overline{\mathrm{LGD}} = \frac{\sum_{t_s} E(t_s,t_s)\,\mathrm{LGD}(t_s)}{\sum_{t_s} E(t_s,t_s)}.$$

The denominator (R2,889.8bn on category 11) counts a rand of default balance once for every TermStep it survives to. It matches the risk suite's weighting and is not a balance-sheet total.

Why South African recovery tails decay

In South African retail and unsecured books the late tail is shaped by law as much as by behaviour. Ordinary debts prescribe after three years under section 11(d) of the Prescription Act 68 of 1969, unless the debtor acknowledges the debt or a summons interrupts prescription. The in duplum rule in section 103(5) of the National Credit Act 34 of 2005 caps the interest that can accrue on a defaulted credit agreement.

Both push recoveries towards a decaying, finite tail. That is the economic case for a fitted decay over an observed tail built on a few long-running accounts. Which of the three shapes fits best remains an empirical question.

Implementation. The app computes the row PVs in `_row_lgds` and the Results table in `_results`. The workbook computes the observed PV and the three tail PVs as single SUMPRODUCT formulas on **Tail_Fit** (columns O to R), and **Results** turns them into LGDs. The two Results tables agree column by column to about 10^{-12}.

10 Stage 7 — LGD to the target TermStep

The LGD-to-Target table needs an LGD for every TermStep up to a target (default 300). Up to the last observed TermStep, each TermStep uses its own row from Stage 6. Beyond it there is no row, so the LGD is *derived* by rolling a base TermStep's extended curve forward (default base 1). The derivation assumes that a loan which has survived to t_s recovers what the base row still had to come.

With $c(b)$ the base row's extended RecoveryPct (observed to its credible bucket, fitted after it):

$$\mathrm{BF}(t_s) = 1 - \sum_{b=\mathrm{Base}}^{t_s-1} c(b), \qquad \mathrm{LGD}_{\mathrm{der}}(t_s) = 1 - \frac{\sum_{b=t_s}^{\mathrm{MaxBucket}} c(b)\, v^b}{v^{\,t_s-1} \cdot \mathrm{BF}(t_s)}.$$

BF is the *balance factor*: the share of the base row's balance still outstanding at t_s. Dividing by it restates the remaining recoveries per rand still owed.

Example 9 — Rolling the base row forward

The base row recovered 0.00086376 in bucket 1, so BF(2) is 0.99913624. The derived LGD at TermStep 2 is 0.9568799, against its own row's 0.9568789, a gap of -1.0×10^{-6}. At TermStep 300, BF is 0.909089.
Because 300 is within the last observed TermStep (301), the final LGD there is the own-row value, 0.9995076.

The table reports the own-row LGD, the derived LGD, the final LGD (own row where it exists, derived beyond), and the gap between own-row and derived values as validation. On category 11 that gap ranges from -6.4×10^{-4} to 1.1×10^{-4}. It also gives the LGD within the short and valuation horizons and the undiscounted recovery still to come. Setting the last observed TermStep below the data's forces derived values earlier, which tests the roll-forward against rows that exist.

Implementation. The app derives the table in `_lgd_to_target` and warns when BF reaches zero or the target runs past MaxBucket. The workbook keeps the base row's running sums on **Base_Row** for **LGD_Target** to read.

11 Design choices and the reasoning behind them

Every stage above rests on a choice that could have been made differently. This section sets out each choice, the reasoning for it, the main alternative, and the evidence on how much it matters. The evidence comes from two books. **Category 11** is long and slow: 59 cohorts from June 2001, 302 TermSteps, and a 25.8% p.a. discount rate. **VB44** is short and fast: 25 cohorts from June 2018 to May 2026, observed to June 2026, 97 TermSteps, R4.02bn opening exposure, and a 17.71% rate. Both were run through the app with the default settings unless stated; Section 13 gives the full sensitivity tables.

Replace the non-credible tail rather than append to it

The model discards observed recoveries beyond the last credible bucket and puts the fitted curve in their place. The alternative is to keep every observed bucket and extend only beyond the last observed one.

The reasoning is that a recovery rate built on one or two accounts measures those accounts, not the portfolio. Keeping it would carry their idiosyncrasy straight into a portfolio LGD. On category 11, buckets 263 to 301 of TermStep 1 rest on one account.

Evidence. Lowering MinExposure to R50m or below keeps all observed buckets on category 11, which is the append-only alternative. The exponential LGD then moves from 0.9816185 to 0.9815978, a difference of 2.1×10^{-5}.

Measure credibility by exposure

The cut is set on the rand exposure behind each recovery rate, because that exposure is the rate's denominator: a rate on a small denominator is volatile. The run-off file also carries AccountCount, and a minimum number of accounts is the natural alternative. The model does not use it, so a large single exposure can pass the cut. A rand cut is also not scale-free, which matters when the book is filtered to fewer vintages. A percentage of opening exposure is available for that reason.

Evidence. MinExposure is the most influential tail judgement on both books. Between R10m and R1bn, the exponential LGD ranges over 6.4×10^{-5} on category 11 and 7.6×10^{-3} on VB44.

One shape for all TermSteps, a level for each

The shape parameters are fitted once, on the reference TermStep, and every TermStep then gets its own level from its own anchor window. Fitting each TermStep

separately is the alternative. It fails in practice: the later TermSteps have short credible tails, and a decay rate fitted on a few points is unstable or undefined. TermStep 1 has the longest and best-populated curve.

The separate level keeps each TermStep's own recovery experience. A loan three years into default recovers at a different rate from a new default, and the anchor window captures that difference.

Evidence. Moving the reference TermStep from 1 to 24 changes the exponential LGD by 1.5×10^{-7} on category 11 and 3.1×10^{-4} on VB44. This shows the shape choice is robust to the reference row. Whether one curve fits every TermStep is a separate question, which the LGD-to-Target validation answers (Section 16).

Anchor the level on a window, not a point

The level is set so the fitted curve matches the observed recoveries in total over the last W credible buckets (12 by default). A single last bucket would be noisy, and monthly collections often have a seasonal pattern that a twelve-month window averages out.

Evidence. Windows of 6 to 36 buckets move the exponential LGD by 8.3×10^{-6} on category 11 and 1.5×10^{-3} on VB44.

Three simple parametric shapes

The tail is extrapolation, so the shapes are deliberately simple, with one or two parameters. Each represents a distinct view of how recoveries die away.

- **Exponential**: a constant monthly decay rate.
- **Power law**: a slowing decay rate, giving a heavy tail.
- **Log-normal**: a rise to a peak and then a fall.

Non-parametric methods, such as chain-ladder development factors, are not used because they cannot extrapolate past the last observed bucket, which is the whole problem. Richer families with more parameters were avoided for the same reason: the extra flexibility is spent fitting noise in the observed range and is unconstrained beyond it. Section 12 sets out how to choose between the three.

Weighted least squares on the recovery scale

The default LM solver fits on the recovery scale, with weights rising with the bucket. This matches the risk suite's own curve fit, so the parameters reconcile to the app's panel.

It also suits the purpose. Fitting in logs (OLS) gives the small late recoveries as much weight as the large early ones, and has to drop zero months. Weighting by bucket concentrates the fit on the later part of the curve, which is the part being extrapolated.

Evidence. The solver choice matters little for the exponential shape: 2.2×10^{-6} on category 11 and 1.2×10^{-4} on VB44. It matters more for the two-parameter and heavy-tailed shapes on the short book: 3.6×10^{-3} for power and 2.6×10^{-3} for log-normal on VB44.

Start the fit at month 24

The first months after default are usually dominated by early arrangements, collections activity and the legal process starting up. Neither of the decaying shapes (exponential or power) can describe that, and fitting through it bends the decay rate. FitStart 24 fits only the settled part of the curve.

Evidence. On category 11 FitStart between 1 and 60 moves the exponential LGD by only 1.0×10^{-6}. On VB44 it matters. Fitting from month 1 lowers the exponential LGD by 2.5×10^{-3} and the power-law LGD by 1.7×10^{-2}, because including the early months flattens the fitted decay and so adds tail recoveries. The default of 24 is the more prudent choice on short books.

A finite horizon of 420 months

Recoveries are run out to MaxBucket, 420 months (35 years) after the cohort start, and set to zero beyond it. A finite horizon reflects that recoveries do not continue indefinitely. In South Africa, prescription after three years and the in duplum rule both bound what can be collected (Section 9). Truncation also errs on the side of a higher LGD, since anything beyond the horizon is excluded rather than assumed.

Evidence. At 25.8% p.a. a rand recovered in month 420 is worth R0.0003 at default; at 17.71% it is worth R0.0033. Moving MaxBucket between 360 and 600 changes the exponential LGD by about 1×10^{-6} on category 11 and by less than 10^{-8} on VB44. Only MaxBucket 302, which barely extends category 11 at all, moves it by more (9.3×10^{-6}). The heavier power-law tail is more exposed to this choice, with a range of 3.1×10^{-5} on category 11 including MaxBucket 302 (8.2×10^{-6} across 360 to 600).

Discount at the file's rate, to the default date

Recoveries are discounted at the rate implied by the risk suite's own discount factors, to the date of default of each TermStep. Using the same rate as the risk suite is what makes the tie-out exact, so the extension changes the recoveries and noth-

ing else. Measuring LGD as a present value at default is the standard construction. Whether that rate is the effective interest rate IFRS 9.B5.5.44 requires is a matter for the ECL methodology; the model accepts any rate typed in.

Evidence. Apart from the choice of vintages, the rate is the most sensitive input, because it reaches every bucket, not just the tail. Between 10% and 20% p.a. the LGD ranges over 5.9×10^{-3} on category 11 and 5.2×10^{-2} on VB44.

No hazard floor by default

The hazard floor sets a minimum monthly recovery rate on the fitted tail. It is set to zero by default because even a small floor is large in total. On TermStep 1 of category 11 a floor of 0.0001 a month would lift the 158 fitted months to at least 0.0158 of undiscounted recovery. That is eighteen times the fitted exponential tail of 0.00089. A floor should only be used where there is evidence of a persistent minimum recovery, such as a portfolio collected on long-running debit orders.

No monotone floor on the extended LGD

The risk suite floors each TermStep's LGD at the highest LGD of any earlier TermStep, which is a way of suppressing noise in the observed tail. The extended LGD is not floored. The extension deals with that noise at its source, by replacing the non-credible recoveries, so a second correction on top would double count. A floor would also make each TermStep's LGD depend on every earlier one. The model reports the risk suite's floor separately (12 TermSteps on category 11), so the effect remains visible.

All vintages by default, with a window tied to the report date

The default uses every cohort, because the oldest cohorts are the only ones that show the late tail at all. Restricting to recent vintages makes the book more current but shortens the observed curve. The extension then has to supply more of the LGD. When a window is used, it is anchored to the report date (Section 4).

Evidence. This is the most consequential choice of all, as Section 13 shows. On VB44 a three-year window gives an exponential LGD of 0.494, against 0.665 from the same cohorts' own observed curve.

Derived LGD beyond the last observed TermStep

Beyond the last observed TermStep there is no row, so the LGD is derived by rolling the base TermStep's extended curve forward. This assumes that a loan surviving to

TermStep t_s behaves like the base row from t_s onwards. It is the same one-curve assumption as the common shape, applied to the timing as well as the decay. The validation column measures it on every TermStep where both values exist.

12 Choosing the tail shape

The three shapes agree inside the data and diverge outside it, which is exactly where the answer is needed. The choice between them is therefore the central judgement in the model. Three kinds of evidence bear on it: how well each shape fits the observed curve, how well it predicts recoveries it was not fitted on, and how much the choice moves the LGD.

12.1 In-sample fit

The LM solver reports the weighted squared error (WSSE) of each shape on the reference row. Lower is better, and the figures are comparable across shapes on the same book.

Book (fit window, points)	Exponential	Power law	Log-normal
Category 11 (24–262, 239)	1.07×10^{-7}	7.68×10^{-7}	1.04×10^{-7}
VB44 (24–89, 66)	1.01×10^{-5}	3.62×10^{-5}	6.15×10^{-6}

The power law fits worst on both books, by a factor of seven on category 11 and between three and six on VB44. The log-normal fits best on both. On category 11 its lead over the exponential is only 3%, but on VB44 it is 39%. The log-normal has a second parameter, so some advantage in fit is expected and does not on its own justify it.

12.2 Out-of-sample backtest

A fairer test is prediction. TermStep 1 was refitted as if its credible data ended earlier, at bucket H, and each shape's tail was then compared with the recoveries actually observed from $H + 1$ to the true last credible bucket. The ratio below is predicted over observed undiscounted recovery in the held-out months. A value of 1.00 is a perfect forecast; above 1 the shape over-predicts recoveries, which understates LGD.

Book, data cut at H	Held-out months	Exponential	Power law	Log-normal
Category 11, $H = 150$	151–262	1.07	1.96	1.33
Category 11, $H = 200$	201–262	0.75	1.15	0.89
VB44, $H = 48$	49–89	1.31	1.95	1.00
VB44, $H = 60$	61–89	0.95	1.44	0.74

The power law over-predicts in all four tests, by between 15% and 96%. Its heavy tail adds recoveries the data did not deliver, so it produces the lowest LGD and is the least prudent of the three. Between the exponential and the log-normal there is no consistent winner. Each is closest in two of the four tests, and each misses by up to about a third in the others. That is the honest scale of uncertainty in any extrapolated recovery tail.

12.3 How much the choice matters

On category 11 the exponential and log-normal LGDs differ by 1.5×10^{-5} (0.9816185 against 0.9816039). On VB44 they differ by 3.0×10^{-4} (0.6112871 against 0.6115920). The power law sits below both: 0.9815814 on category 11 and 0.5998533 on VB44, 1.1 LGD points under the exponential.

Example 10 — Why the shapes diverge only in the tail

On TermStep 1 of category 11, the observed recoveries to bucket 262 total 0.0904 of the defaulted balance, undiscounted. The three fitted tails add 0.00089 (exponential), 0.00135 (log-normal) and 0.00215 (power law).
The shapes disagree by a factor of 2.4 in the tail, but the tail is only 1% to 2.3% of the total recovery, and at 25.8% p.a. it is discounted to a fraction of that.

12.4 Recommended selection rule

On this evidence the selection rule below is proposed:

1. **Use the exponential shape as the default.** It has one parameter, so it is always identified. It is the risk suite's own default, it fits within a few per cent of the best shape on a long book, and it is not the least prudent of the three.
2. **Move to the log-normal** only where its fitted peak ($e^{\mu-\sigma^2}$) lies inside the fit window, its WSSE is materially lower, and a holdout backtest on the same book supports it. On both books tested the peak lies before FitStart (23.7 months on category 11, 20.7 on VB44), so the rule keeps the exponential on both.
3. **Do not use the power law** as the selected shape. Report it as the optimistic bound of the range.

4. **Always report all three.** The spread between them is the model's own measure of tail uncertainty and should accompany the selected result.

13 Sensitivity and materiality

13.1 How much of the LGD the tail carries

The first question for any extrapolation is how much rests on it. The table shows the share of PV recoveries supplied by the fitted tail, exposure-weighted across TermSteps.

Book	Exponential	Power law	Log-normal
Category 11: tail share of PV recoveries	0.30%	0.50%	0.38%
tail PV as a share of exposure	0.006%	0.009%	0.007%
VB44: tail share of PV recoveries	2.2%	5.0%	2.1%
tail PV as a share of exposure	0.84%	1.98%	0.81%

On category 11 the tail is immaterial. The book recovers little in total, the credible curve already runs to month 262, and at 25.8% p.a. a rand at month 262 is worth R0.007 at default. On VB44 the tail carries 0.8 LGD points under the exponential shape and 2.0 points under the power law. The shorter the credible curve and the lower the rate, the more the result depends on the extension. The judgements in Section 11 matter in proportion.

13.2 One-at-a-time sensitivity of the tail judgements

Each judgement was moved across the values shown with everything else at its default. The table gives the range of the exposure-weighted LGD (exponential shape) across those values.

Judgement	Values tested	Category 11	VB44
MinExposure	R10m, R25m, R50m, R100m, R250m, R500m, R1bn	6.4×10^{-5}	7.6×10^{-3}
FitStart	1, 6, 12, 24, 36, 60	1.0×10^{-6}	4.1×10^{-3}
Window W	6, 12, 24, 36	8.3×10^{-6}	1.5×10^{-3}
Reference TermStep	1, 6, 12, 24	1.5×10^{-7}	3.1×10^{-4}
Solver	LM, OLS	2.2×10^{-6}	1.2×10^{-4}
MaxBucket	302, 360, 420, 480, 600	9.4×10^{-6}	$< 10^{-8}$
Tail shape	exponential, power, log-normal	3.7×10^{-5}	1.2×10^{-2}

On the long book no tail judgement moves the LGD by as much as one hundredth

of a point. On the short book MinExposure, FitStart and the shape each move it by between 0.4 and 1.2 points. On VB44 the power law is the most sensitive shape to MinExposure, FitStart, the window, the solver and MaxBucket; its range under MinExposure alone is 2.6×10^{-2}.

13.3 Sensitivity of the data and basis choices

Two inputs are not tail judgements but move the LGD far more: the discount rate, and the vintages used.

Setting (exponential shape)	Category 11	VB44
Discount rate implied by the file (25.8%; 17.71%)	0.9816185	0.6112871
Discount rate 20% p.a.	0.9794050	0.6218171
Discount rate 15% p.a.	0.9768996	0.5978143
Discount rate 10% p.a.	0.9735367	0.5695617
All vintages	0.9816185	0.6112871
Last 10 years (own observed LGD)	0.9708041 (0.9725203)	= all vintages
Last 5 years (own observed LGD)	0.9595034 (0.9718202)	0.5605452 (0.5969350)
Last 3 years (own observed LGD)	undefined (0.9700675)	0.4935694 (0.6649416)

The vintage rows show the extended LGD against the LGD of the same cohorts' own observed curve. A shorter window shortens the curve, so the observed LGD omits more of the recoveries still to come, and the extension has more to add. On VB44 the extension lowers the three-year LGD by 17 points, and the shapes disagree by 18 points (0.392 to 0.575). At that point the tail is no longer an adjustment; it is most of the answer. On category 11 three years of data give a rising curve, so the exponential and power-law shapes are undefined and only the log-normal (0.9688071) is available.

14 What each implementation produces

14.1 The risk suite app

The app organises work into *projects*. A project holds debug zips (one per category or portfolio), *scenarios* (parameter sets, with optional per-zip overrides) and their results. Running a scenario applies the model to every zip and stores, for each pair:

- the headline figures: exposure-weighted LGD extended and original, the uplift, TermStep 1 LGD and the horizon LGDs;

- the Results, LGD-to-Target and tail-fit tables;
- the forecast-curve panel: fit window, $\lambda, \gamma, \mu, \sigma$ fitted and used, and the three levels on TermStep 1;
- the effective configuration (report month, vintages used, rate, credibility) and every warning raised.

A matrix view compares zips and scenarios. The client's own applied recovery curves can be uploaded for comparison on charts, but they do not feed the extension. Exports include a values workbook and a formula-driven workbook for one zip and scenario, a results bundle across zips, a curves workbook and a project summary.

14.2 The validation workbook

The workbook runs one event of one debug zip at a time. Its **Control** sheet shows the same headline figures and parameters as the app's result screen. Control also shows the vintages used, the exposure included and the discount rate, plus eight checks that must all pass before the status reads READY:

- the vintage choice is valid;
- the input rows are contiguous;
- MaxBucket and Target are within the 420 columns built;
- the replica ties to the file;
- the balance factor at the target is positive;
- λ and γ are defined;
- the LM fit is up to date;
- the run-off self-check passes.

A compare-with-app block takes the app's eight headline figures and flags any difference above 10^{-6}. Nine sheets are visible: Control, Config, Input, Obs_R, Tail_Fit, Fit_OLS, Fit_LM, Results and LGD_Target. The data sheets and intermediate grids (Runoff_Input, Obs_Index, Obs_LGD, Obs_E, Shapes, Base_Row, Lists, Dash_Data) are hidden.

15 Keeping the app and the workbook in step

The two implementations share no code. The app is numpy and sums cohort by cohort; the workbook is SUMIFS and SUMPRODUCT. Agreement between them is therefore evidence that both are right, not that one copies the other.

They were reconciled on the files supplied in October 2026. On category 11 with the default settings, the workbook reproduces the app's forecast-curve panel to the digits shown. All 302 Results rows and 300 LGD-to-Target rows agree to about 10^{-12}. The report-date logic was then exercised on both books under a set of test settings: OLS, log-normal, MinExposure R10,000,000, window 24, FitStart 6 and short horizon 36.

Book	Vintages	Window used	Cohorts	LGD	Gap
Category 11	All	to 2026-07	59 / 59	0.9815915	$< 10^{-14}$
Category 11	Last 5 years	2021-08–2026-07	17 / 59	0.9586674	$< 10^{-14}$
VB22	All	to 2026-07	71 / 71	0.8632228	$< 10^{-14}$
VB22	Last 5 years	2021-08–2026-07	34 / 71	0.8293220	$< 10^{-14}$
VB22	5 years to 2026-06	2021-07–2026-06	35 / 71	0.8445961	$< 10^{-14}$
Category 11, default settings	All	to 2026-07	59 / 59	0.9816185	1×10^{-12}

The few differences between the two are by design:

- **Scope.** The app runs many zips and scenarios; the workbook runs one event at a time.
- **Vintage start month.** Only the app can start from a given month; the workbook offers "last N years" only.
- **Run-off layout.** The workbook's SUMIFS need the StructOK layout; the app does not.
- **Size.** The workbook is built for 302 TermSteps and 420 buckets; the app sizes itself to the file.
- **LM starts.** Different starting points, same bounds, same minimum.

For any new file, run the app, paste its eight headline figures into the workbook's compare block, and confirm every line matches.

16 Known weaknesses, controls and monitoring

16.1 Known weaknesses

- **The tail is an assumption.** Beyond the last credible bucket the recoveries are the chosen shape, not data. The backtest shows that even the better shapes can miss held-out recoveries by up to a third. The model's answer to this is to report all three shapes and their spread, not to claim precision.
- **Short histories.** Where few vintages reach the late buckets, the extension sup-

plies most of the LGD and the shapes diverge (VB44, three years: 0.392 to 0.575). Results on such books need explicit judgement and should not be used on the model's output alone.

- **One curve for all TermSteps.** The shape comes from the reference row, and the derived LGD from rolling the base row forward. The validation column tests this wherever a TermStep has both an own-row and a derived LGD. On category 11 the two agree to within 6.4×10^{-4} everywhere. On VB44 they do not. The own-row LGD is 0.5 to 1.6 points below the derived value at TermSteps 27 to 48, which carry R0.55bn to R1.1bn of exposure each. From TermStep 78 it is 5 to 7 points below. Loans that have been in default longer recover more on that book than TermStep 1's curve implies, so the derived LGD beyond the last observed TermStep (96) is likely overstated there. That errs on the prudent side, but it should be tested by moving the base TermStep before the derived values are relied on.
- **Thin late TermSteps keep their own rows.** Up to the last observed TermStep the final LGD is the TermStep's own row, however little exposure is behind it. On VB44, TermSteps 90 to 96 rest on about R6m each, and TermStep 91's own-row LGD is 0.907 against 0.978 derived. Setting the last observed TermStep lower switches such rows to the derived value in the LGD-to-Target table; the exposure-weighted headline LGD, which uses the Results rows, does not change.
- **Recovery is a balance movement.** Any non-cash change in the run-off balance is treated as recovery (Section 2). The quality of the run-off file is therefore a model input.
- **Log-normal identifiability.** On short fit windows the log-normal peak lies outside the data and μ, σ are poorly determined. On the five-year category 11 book the fit stops at the $\mu = 30$ bound, and the app does not flag this automatically; a parameter at its bound should be checked by hand.
- **A rand MinExposure is not scale-free.** The same cut means different things on a large and a small book. Under a vintage filter, the app warns when a rand cut leaves fewer credible buckets on the reference row than the window.
- **Static and historical.** There is no forward-looking adjustment, and the discount rate is taken from the file. Both are matters for the ECL methodology.

16.2 Controls built into the model

- **Tie-out before extension.** The rebuilt triangle is checked against the risk suite's exposure, cash and LGD. The workbook will not show READY unless they agree to within 10^{-9}. The app does not stop the run; it disables a vintage filter when the run-off rebuild fails its check, and warns when the replica LGD misses the file by more than 10^{-6}.

- **Undefined is undefined.** A shape that does not decay, a log-normal with no peak (OLS), or a failed fit gives a blank and a warning, never a number; if it is the selected shape, the run stops with an error.
- **Bounded, multi-start fitting.** Both LM implementations (the app and the workbook macro) search within the same bounds from several starting points and reject steps outside them. A poor start is therefore unlikely to strand the fit in a local minimum and cannot cause an overflow.
- **Warnings on settings.** The app flags thin credibility under a vintage filter, a MaxBucket below the data, a target beyond MaxBucket, and a balance factor that reaches zero.
- **Two independent implementations.** The app (cohort sums in numpy) and the workbook (SUMIFS in Excel) share no code and agree to about 10^{-12}. A coding error would have to be made twice, identically, to go unseen.

16.3 Recommended monitoring

At each refresh of the debug files:

1. Re-run the holdout backtest of Section 12 on each book and record the predicted-to-observed ratio for every shape.
2. Track λ, γ, μ, σ, the last credible bucket of the reference row and the fit window over time. A large movement between refreshes means the data have changed materially and should be explained.
3. Report the selected LGD with the three-shape spread and the share of PV recoveries carried by the tail.
4. Reconcile at least one book between the app and the workbook using the compare-with-app block.
5. Review MinExposure against the size of each book, and prefer the percentage basis where vintage windows are used.

IFRS 9 — time value of money (paraphrased)

IFRS 9.5.5.17(b) requires expected credit losses to reflect the time value of money. IFRS 9.B5.5.44 requires them to be discounted to the reporting date at the effective interest rate determined at initial recognition, or an approximation of it. In this model the time value enters through $v = (1 + r)^{-1/12}$ applied bucket by bucket. The rate r is taken from the risk suite's file unless overridden.

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